

NoboGhat

Software Project Proposal & System Design Document (Version 1.0)

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Purpose: Foundation document for the NoboGhat project.

1. Executive Summary

NoboGhat is a Bangladesh-focused inland waterway logistics platform that enables small traders, farmers, and boat owners to share cargo space, discover routes, and book transport digitally. The project aims to modernize traditional river transport while remaining simple enough to evolve from a university project into a production system.

2. Background

Bangladesh has one of the world's largest inland waterway networks, yet much of the booking process remains manual. Small businesses often rely on brokers, face uncertain pricing, and cannot efficiently share boat capacity.

3. Problem Statement

Current cargo transport suffers from fragmented communication, limited visibility, underutilized boats, and a lack of digital booking tools for small shipments.

4. Proposed Solution

NoboGhat will provide user registration, boat listings, shared cargo booking, route discovery, trip scheduling, booking management, and administrative monitoring through a web application.

5. Vision

To become Bangladesh's trusted digital platform for inland water logistics and shared cargo transport.

6. Mission

Reduce transport barriers for rural communities through accessible technology.

7. Objectives

- Digital booking
- Shared cargo optimization
- Transparent scheduling
- Better utilization of inland waterways

8. Stakeholders

Farmers, Traders, Boat Owners, Transport Coordinators, Administrators.

9. MVP Scope

Authentication, boat listing, route management, cargo booking, capacity validation, booking history, admin dashboard.

10. Core Modules

Authentication, User Management, Boat Management, Route Management, Cargo Booking, Dashboard, Notifications (future).

11. User Journey

Register → Browse Routes → Select Boat → Book Cargo → Receive Confirmation → Delivery → Review.

12. System Architecture

Frontend (HTML/CSS/JS) → Backend API (Java Spring Boot planned) → Database (MySQL planned).

13. OOP Mapping

User, Farmer, BoatOwner, Admin, Boat, Cargo, Route, Trip, Booking, Notification. Demonstrates encapsulation, inheritance, abstraction, interfaces, and associations.

14. Technology Roadmap

Phase 1: HTML/CSS/JS

Phase 2: Java Backend (Spring Boot)

Phase 3: MySQL

Phase 4: REST APIs

Phase 5: React (future)

Phase 6: Mobile App (future).

15. Development Plan

Planning → UI → Frontend Logic → Backend → Database → Authentication → Testing → Deployment.

16. Risks

Connectivity, seasonal river conditions, adoption, data quality, regulatory considerations.

17. Future Enhancements

GPS tracking, weather integration, flood alerts, digital payments, warehouse support, analytics.

18. Conclusion

NoboGhat is designed as both an academic capstone-quality project and a long-term portfolio/startup foundation. The document serves as the blueprint for future implementation.